MOLECULE: Molecular-dynamics and Optimized deep Learning for Entropy-regularized Classification and Uncertainty-aware Ligand Evaluation

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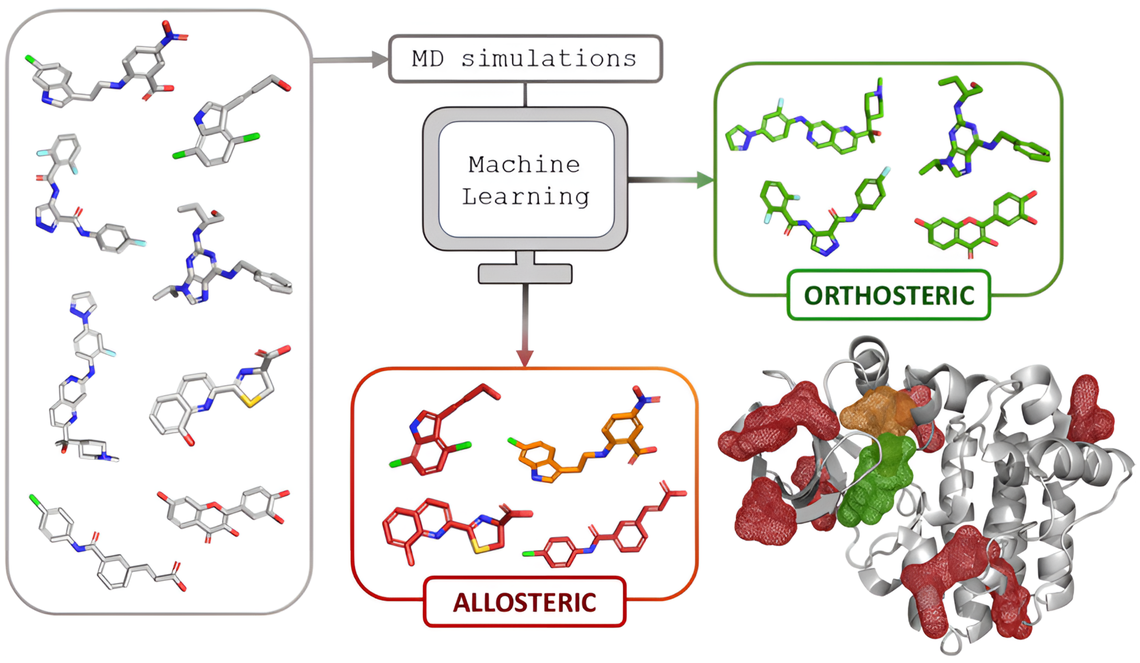
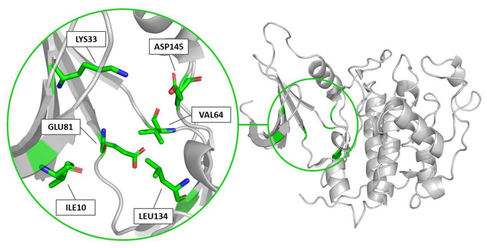
Graphical abstract



MD simulations & ligand chemical structure descriptors



Selection of six conserved residues among all studied structures that describe the ATP binding site.

To take the dynamic response of the protein active sites to ligand binding into account, we monitored a number of descriptors related to the modulation of protein functional motions.

- **Root-mean square deviations** of the positions of the selected residues during the simulations with respect to the initial frame.
- **Distances** between each pair of the selected residues.
- **Pair-distance fluctuations**, which is an indicator of the coordination of motions between all residues during the simulations.
- **Morgan fingerprint** is a type of molecular fingerprint used in cheminformatics to represent the structure of a molecule in a way that is useful for machine learning and similarity searches. It is based on the Extended-Connectivity Fingerprint (ECFP) algorithm, which encodes molecular structures as binary vectors.

Dataset construction

- **Structures**: 139 allosteric and 140 orthosteric
- **MD simulations**: 4 replicas of 100 ns each
- **Atom sets**: "all atoms" and "backbone" each replica

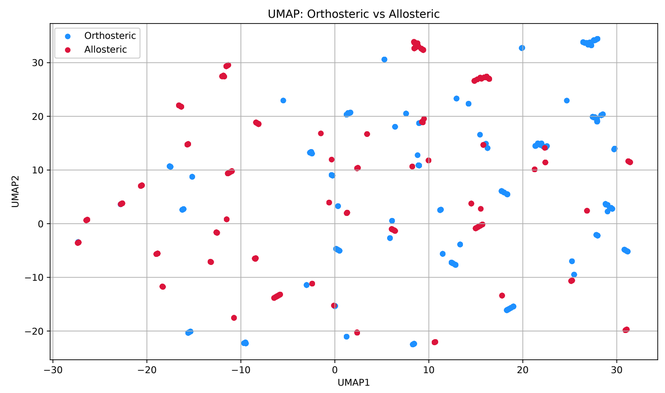
Original Dataset
279 data x 130048 features

Feature engineering + Data augmentation

Dataset complete

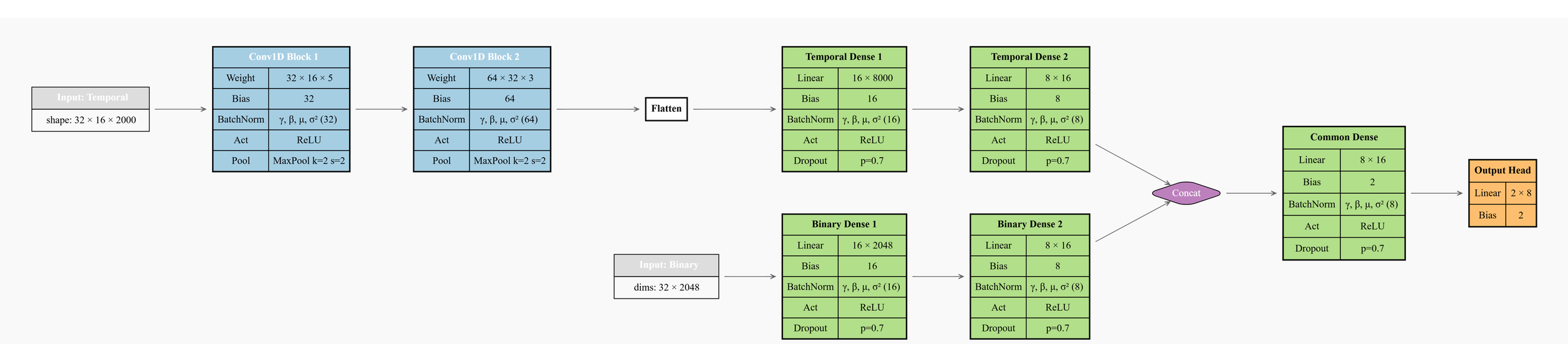
2064 data x	Dyn. part	Bin. part
	16 x 2000	2048

Dataset distribution



The figure displays a 2D UMAP embedding of the data points, where each point corresponds to a single sample. Samples labeled as orthosteric are shown in blue, while allosteric samples are shown in red. The embedding reveals the overall distribution and overlap between the two classes in the reduced latent space.

Neural Network architecture



Method

Dynamic entropy-based regularized Focal Loss

$$\mathcal{L} = FL(y, \hat{y}; \gamma) + \lambda \cdot H(\hat{y})$$

- $\hat{y}$ represents the **predicted probability** for a given class,
- $FL(y, \hat{y}; \gamma) = -y(1-\hat{y})^{\gamma}\log(\hat{y}) - (1-y)\hat{y}^{\gamma}\log(1-\hat{y})$ is a **dynamic focal loss** term, with focusing parameter scheduled by $\gamma = \min(2, 0.2 \cdot epoch)$,
- $H(\hat{y}) = \sum \hat{y}\log(\hat{y})$ is a **entropy-based regularization** term

AdaBelief Optimizer

$$\theta_t = \underset{x \in \mathcal{F}}{\operatorname{argmin}} \left\| \sqrt[4]{\hat{s}_t}\left(x - \left(\theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{s}_t} + \epsilon}\right)\right)\right\|$$

- $g_t = \nabla_\theta \mathcal{L}_t(\theta_{t-1})$ is the **gradient** of the loss function,
- $m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$ is the **exponential moving average** of g_t,
- $s_t = \beta_2 s_{t-1} + (1-\beta_2)(g_t - m_t)^2 + 10^{-16}$ is the **exponential moving average** of $(g_t - m_t)^2$,
- $\hat{m}_t = \frac{m_t}{1-\beta_1^t}$ is the **bias correction** of m_t,
- $\hat{s}_t = \frac{s_t}{1-\beta_2^t}$ is the **bias correction** of s_t,
- $\beta_1 = 0.9$, $\beta_1 2 = 0.999$ are **smoothing parameters**,
- $\alpha = 10^{-3}$ is the **learning rate**

Threshold-based classification strategy

We introduce an additional "**uncertain**" class for cases where the highest **prediction probability** is below a certain threshold, ensuring more reliable decisions.

Training setting

The model is trained for **200 epochs** using **mini-batches** of size **32**. To ensure training stability and control model capacity, we apply **max-norm regularization** with a threshold of **3.0** to the weights of all convolutional and fully connected layers.

Feature importance

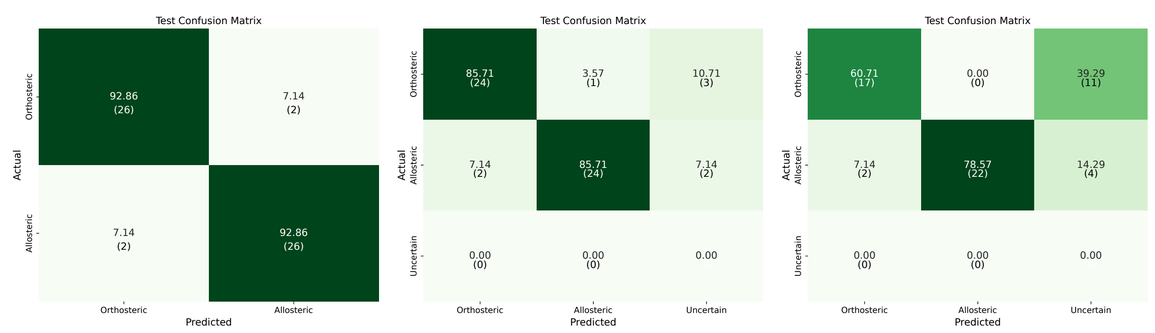
Type	Orthosteric	Allosteric	Average
Tot Dynamics	90.45%	90.62%	90.53%
Tot Fingerprint	9.55%	9.38%	9.47%
Avg Dynamics	37.73%	38.21%	37.97%
Avg Fingerprint	62.27%	61.79%	62.03%

Computed by **Integrated Gradients**

Metrics

Metric	Th. = 0.50	Th. = 0.80	Th. = 0.95
Accuracy	92.86%	85.71%	69.64%
F1 score	92.86%	89.73%	80.17%
Cohen's Kappa	0.86	0.74	0.52

Results



Threshold = 0.50 Threshold = 0.80 Threshold = 0.95

Conclusion

The proposed model effectively distinguishes orthosteric and allosteric inhibitors by integrating dynamical and fingerprint information. Notably, with a **threshold of 0.80**, it achieves a **strong balance** between **performance** and **confidence**, reaching **85.71% accuracy**, **89.73% F1-score**, and a **Cohen's kappa** of **0.74**. **Feature importance** analysis highlights the **dominant role** of **dynamical features**—contributing **over 90%** of total attributions—demonstrating their critical influence on model decisions.

References

[1] E. Frasnetti, I. Cucchi, S. A. Serapian, L. F. Pavarino, G. Colombo. MOLECULE: Molecular-dynamics and Optimized deep Learning for Entropy-regularized Classification and Uncertainty-aware Ligand Evaluation, *Manuscript in preparation*.